E-Commerce Sales Intelligence & Customer Insights

Case Study — Capital One Data Analyst Challenge

Venkata Lakshmi Parimala | 2026

1. What I Did

This is a run against a Capital One data analyst challenge: two years of e-commerce data (sales transactions, buyer profiles, and 11 separate vendor catalog files), scoped around three strategic decisions — whether to maintain 24/7 staffing, whether to continue a wide product catalog strategy, and whether to continue the referral discount program — plus five specific business questions covering staffing, segment profitability, the referral program, Black Friday, and where to focus acquisition spend next.

Honestly, most of the actual work here wasn't the analysis, it was getting the data into a state I could trust. All three source files had real problems (a silent join failure, mixed date formats, a corrupted sentinel value, inconsistent margin scales across vendors), and none of them announced themselves. I only caught the worst one, 46 SKUs silently dropping out of the vendor join because one file used lowercase keys, by checking the match rate before and after cleaning instead of assuming the join worked because it didn't error.

2. Technical Implementation

2.1 Data Processing Pipeline

Loading:

Ingested three source types — sales.csv, buyer.csv, and 11 vendor CSV files stacked and reconciled into a unified catalog. Each vendor file had structural differences requiring file-aware handling before concatenation.

Cleaning:

Applied targeted fixes to each dataset before joining. All cleaning logic is isolated in src/clean.py; all thresholds and constants live in src/config.py so nothing is hardcoded in the analysis.

Joining:

Merged sales → buyers on buyer_id and sales → vendors on sku_id. SKU match rate verified at 100% after cleaning (was 991/1,037 ≈ 95.6% before the casing fix) to confirm no silent data loss.

Feature Engineering:

Built derived columns in src/features.py: timezone-converted timestamps (buyer local time → Central), Black Friday flag, revenue/profit eligibility, quarterly labels, and referral cost estimate. All features are computed deterministically from config values.

Checks:

src/checks.py runs 27 assertions across all five pipeline stages. Any silent failure (wrong dtype, empty join, missing column) raises immediately with the exact value that failed. A companion pytest suite (6 tests) additionally guards against unverified sourcing claims and documentation drifting out of sync with the code — including two tests that re-run the live pipeline to confirm README's numbers still match reality, rather than trusting a hardcoded value.

Outlier Detection:

Two numeric fields were checked for outliers and handled explicitly rather than silently dropped: quantity values above a 50-unit-per-line-item threshold (including a corrupted sentinel value of ~327,783,095) were flagged via quantity_is_valid and excluded from quantity-dependent metrics; negative profit

margins (111 rows across 7 vendor files) were flagged via profit_margin_is_neg and retained, since a negative margin is a plausible real business condition (loss-leader pricing) rather than a data error.

2.2 Data Quality Issues Fixed

Sales (sales.csv):

- buyer_id whitespace padding on ~18,000 rows — caused silent join failures before stripping
- ~24,000 exact duplicate rows removed
- order_status typo: "Deliverred" → "Delivered" (3,714 rows)
- order_datetime stored in three mixed formats in one column: m/d/yy H:MM, d-Mon-yy (date-only), and 10-digit Unix epoch (~4,665 rows)
- Unix epoch zero sentinel "1/1/70 0:00" (~190 rows) nulled out before parsing
- ~340 rows with null buyer_id dropped (unjoinable)

Buyers (buyer.csv):

- 2,634 null buyer_id rows dropped
- Duplicate buyer records deduped to one row per buyer
- is_referred stored with four different boolean spellings across rows, normalized to real Python bool
- Missing region and timezone imputed from state using an empirical lookup, preserved the company's non-standard "Other" region bucket

Vendors (11 files):

- NKDLI.csv stored all key join columns in lowercase, caused 46 SKUs to silently drop from the join. Fixed by uppercasing before sku_id construction
- ROPIK.csv and JDTOJ.csv stored profit_margin on a 0–100 scale; all others used 0–1 fraction. Normalized by dividing only those two files by 100
- ROPIK.csv had a currency symbol in the price column ($7.9), stripped before numeric cast
- Boolean columns inconsistent across files, normalized with a shared mapping

2.3 Key Assumptions

This version of the challenge came with an explicit assumptions document. A1, A4, A5, A7, A8, and A9 below are stated facts from that document, not inferences — an earlier draft of this analysis had incorrectly hedged several of these as unconfirmed placeholders before the source document was located. A2, A3, A6, and A10 remain genuine design choices this analysis made on its own, since the source document doesn't specify them.

#	Assumption	Source	Impact if Wrong
A1	Timestamp of purchase is based on the customer's timezone, converted to Central for Q1	Given (assumptions doc)	This is now a stated fact rather than an inference — the Q1 finding rests on solid ground
A2	Buyers with null/unresolved timezone default to Central	Design choice	Minor, very few remain unresolved after imputation
A3	"Customer segment" = customer_group (VIP/Frequent/Regular/Occasional/Do	Design choice	Changes Q2/Q4/Q5 groupings

#	Assumption	Source	Impact if Wrong
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A4	Referred customers receive 10% off their first order	Given (assumptions doc)	Dollar estimate is now a confirmed calculation, not a placeholder — the core behavioral finding (no lift) held under both the old and new figures
A5	Black Friday eligibility uses the customer's local timezone (per the assumptions doc)	Given (assumptions doc)	Consistent with A1's timezone convention
A6	Only Delivered orders count toward revenue/profit; all statuses count toward volume	Design choice	Q1 uses full order count; Q2–Q5 use delivered-only
A7	Customers pay tiered shipping: <$50 → $7.99, $50–$99.99 → $4.99, ≥$100 → free	Given (assumptions doc)	Now modeled into order_revenue, previously omitted entirely
A8	The company pays a flat $4.99/order to its carrier, regardless of size or weight	Given (assumptions doc)	Now modeled into order_profit as carrier_cost, previously omitted entirely
A9	Taxes are ignored — pre-calculated into the raw price data, per the assumptions document	Given (assumptions doc)	No adjustment needed; figures are already net of tax
A10	The 10% referral discount applies to the shipping-inclusive order total, not just the product subtotal	Design choice	The assumptions doc doesn't specify this either way; this reading adds roughly $2,200 to the Q3 estimate vs. a product-only reading — the core finding (no behavioral lift) holds either way

3. Analysis & Findings

3.1 Q1 — Staffing: Is 24/7 Coverage Justified?

Finding:

Not really, no. Only 8.5% of orders land in the overnight window (12am–6am Central). That number alone doesn't say much until you compare it to something, if orders were spread evenly across the day, a 6-hour window would still capture 25% of volume. 8.5% against that baseline is the actual case for cutting overnight staffing, not the raw percentage on its own.

Recommendation:

I'd move off flat 24/7 coverage and toward a tiered model, full staffing during the peak window, a smaller on-call crew overnight. One caveat before acting on this: fulfillment SLA and cancellation rate should be tracked as guardrails, since this finding only tells you demand is low overnight, not that service quality won't suffer if coverage drops.

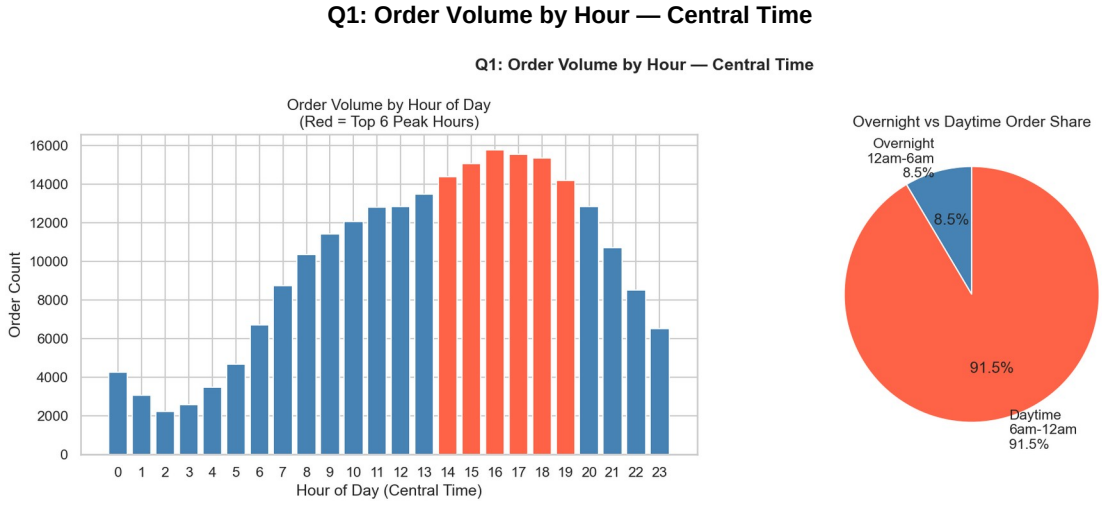


Figure 1: Order volume by hour of day (Central time)

3.2 Q2 — Profitability: Which Segments Drive Profit?

Finding:

VIP customers come out on top every single quarter, no surprises there. Profit per order here now includes the shipping margin (what customers pay minus the flat $4.99 carrier cost) and the Black Friday discount effect, so these numbers shifted slightly from an earlier pass that didn't model either. What I didn't expect going in was the Dormant segment: about 23,700 of the ~47,500 buyers, roughly half, have zero orders in the entire two-year window. Not low activity, zero. That's a bigger story than the segment ranking itself.

Recommendation:

Retention spend should go toward VIP and Frequent, since they're the most stable profit drivers quarter over quarter. But before doing anything about the Dormant group, I'd check whether their signups skew recent, if so, "Dormant" is probably just a default label assigned at signup, not a sign that half the customer base churned.

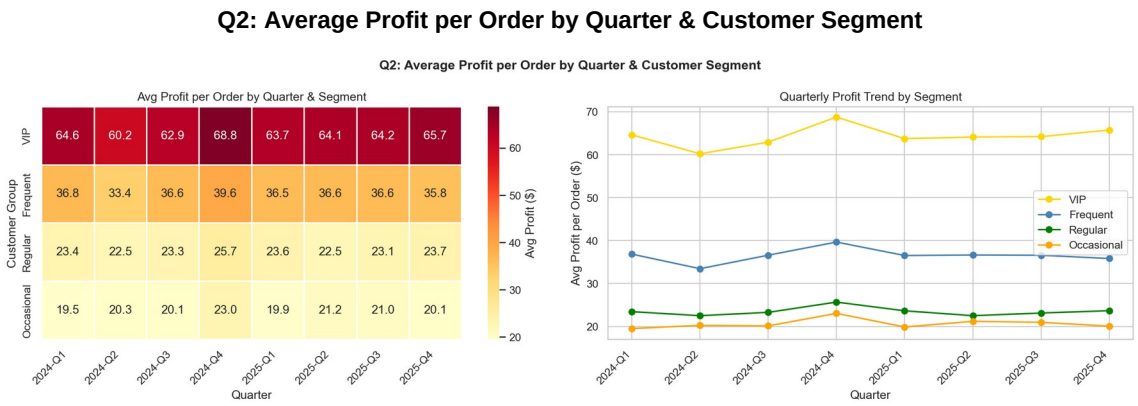


Figure 2: Average profit per order by quarter and customer segment

3.3 Q3 — Referral Program: What Is It Costing?

Finding:

This one surprised me a little. I expected referred buyers to look at least somewhat different from organic ones, instead, revenue per buyer ($788 vs. $799) and orders per buyer (9.46 vs. 9.40) are nearly identical. Whatever the referral discount is doing, it doesn't show up in buyer behavior. The 10% rate itself is a stated fact from the challenge's assumptions document, not a guess, so the ~$24,368 cost estimate is a real calculation now, not a placeholder. One judgment call worth flagging: this figure applies the 10% to the shipping-inclusive order total, since the assumptions document says "10% off first order" without specifying whether that means product cost alone or everything the customer pays.

Recommendation:

The core finding, no measurable behavioral lift from being referred, doesn't depend on that shipping question, so it's solid either way. I'd want a 90-day cohort retention read on referred vs. organic buyers next, and would consider restructuring the incentive itself: maybe a smaller first-order discount plus a credit on the second order, so the program creates an actual retention signal instead of a one-time discount nobody seems to respond to differently.

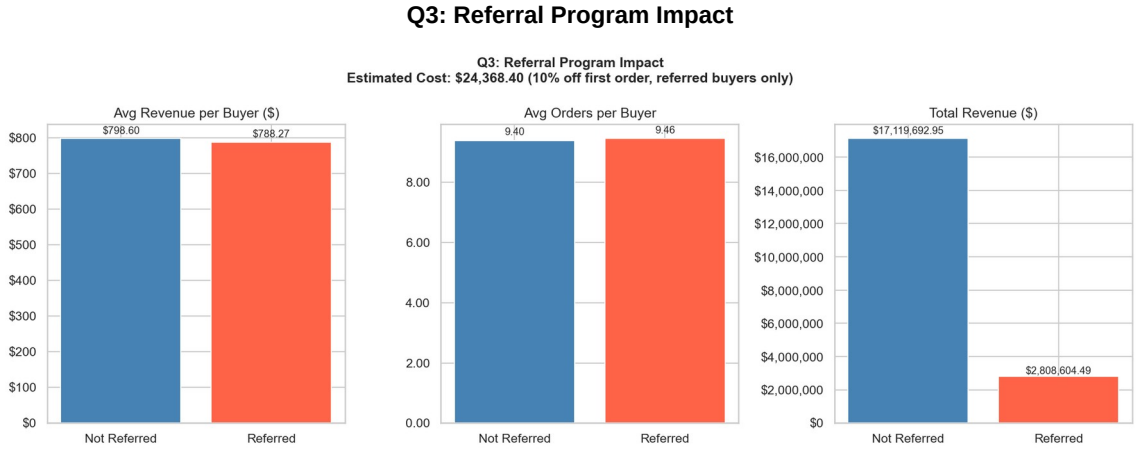


Figure 3: Referred vs. non-referred buyer comparison — estimated cost: $24,368.40 (10% off first order, referred buyers only)

3.4 Q4 — Black Friday: Volume Spike or Revenue Event?

Finding:

There's a real spike, but it's smaller than I think most people would guess, and smaller still once the 20% Black Friday discount is actually applied (it isn't reflected in the raw price data, so it has to be modeled explicitly, per the assumptions document). About 1.1% of orders and roughly 1.0–1.3% of revenue for the year land on that single day. The more interesting cut is by segment: Occasional buyers show the biggest relative pull toward Black Friday of anyone, which reads less like "BF just moves VIP purchases around" and more like it's genuinely pulling in shoppers who otherwise wouldn't have ordered.

Recommendation:

I'd treat Black Friday as an acquisition play aimed at Occasional buyers rather than a revenue event — it's not big enough to matter as the latter, especially once the real discount is priced in. Worth following up with those buyers in Q1 to see if any convert toward Frequent, and tracking BF's revenue share year over year in case it starts cannibalizing regular-season purchases instead of adding incremental ones.

Q4: Black Friday Impact on Annual Sales by Segment

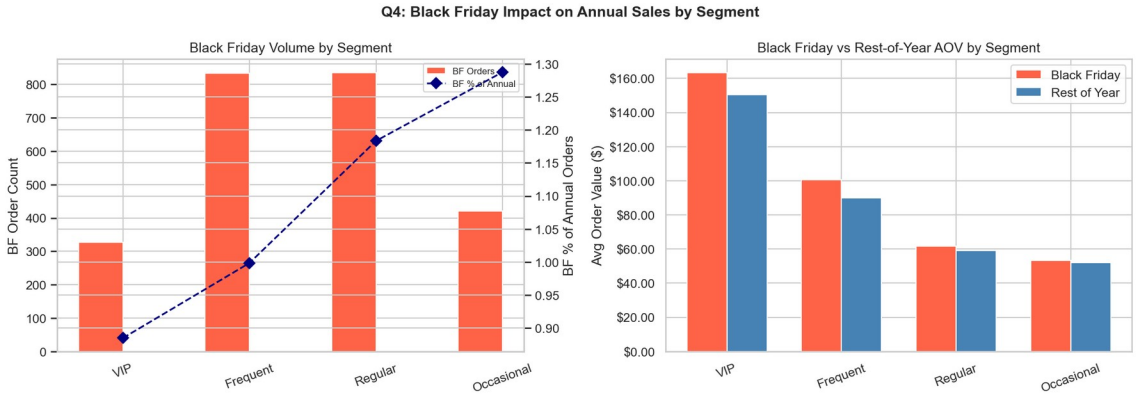


Figure 4: Black Friday volume and average order value vs. rest-of-year, by segment

3.5 Q5 — Best Customers: Where Are They?

Finding:

Geography turned out to be less interesting than I expected, revenue per buyer only ranges from about $791 to $802 across regions, which is close enough that I wouldn't lead an acquisition strategy with "target this region." Segment matters far more: VIP buyers order roughly 26 times over the window, versus under 5 for Occasional buyers, that gap is where the real signal is.

Recommendation:

I'd put acquisition spend behind channel quality rather than geography, since region just isn't a strong lever here. For ongoing tracking, six KPIs cover the gaps this dataset currently leaves open: buyer activation rate, revenue per new buyer by region, 90-day repurchase rate, BF new-buyer conversion rate, catalog profit margin %, and referred-buyer 90-day retention.

Q5: Best Customer Base & Acquisition Opportunity

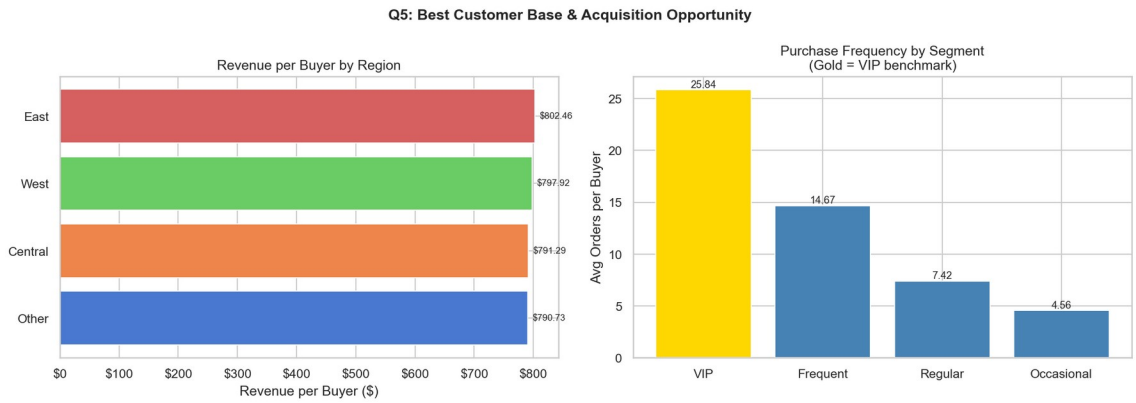


Figure 5: Revenue per buyer by region and purchase frequency by segment

3.6 Catalog Strategy — Is a Wide Assortment Still the Right Call?

Finding:

This is the third strategic decision the brief asks about, and it doesn't have a clean yes/no answer the way staffing does. Every one of the 1,037 SKUs in the catalog has sold at least once, so breadth by itself isn't creating obvious dead inventory. But revenue is heavily concentrated: the top 20% of SKUs generate two-thirds of all revenue, while the bottom half of the catalog contributes under 10%. That's a fairly typical long-tail pattern, not a red flag on its own.

What did concern me is vendor continuity: only 47% of SKUs (487 of 1,037) are tied to a vendor still marked currently active. More than half the catalog could lose its supply source without any warning baked into the data. And loss-leader SKUs (111, or 10.7% of the catalog) aren't spread evenly, they're concentrated in Pet Supplies (26% of its SKUs are negative-margin, average margin just 5.1%), plus Toys & Games, Sports & Outdoors, and Books. Electronics and Beauty, by contrast, are carrying margins north of 50%.

Recommendation:

I wouldn't narrow the catalog on breadth alone, the long tail is still selling, so cutting it risks losing real revenue for a data-quality problem that doesn't actually exist. Instead I'd prioritize two things: re-sourcing or replacing the roughly 550 SKUs tied to inactive vendors before that turns into a stockout problem, and reviewing Pet Supplies specifically, since it's the one category losing money on a quarter of its SKUs rather than carrying the occasional loss-leader.

Catalog Strategy: Revenue Concentration & Category Health

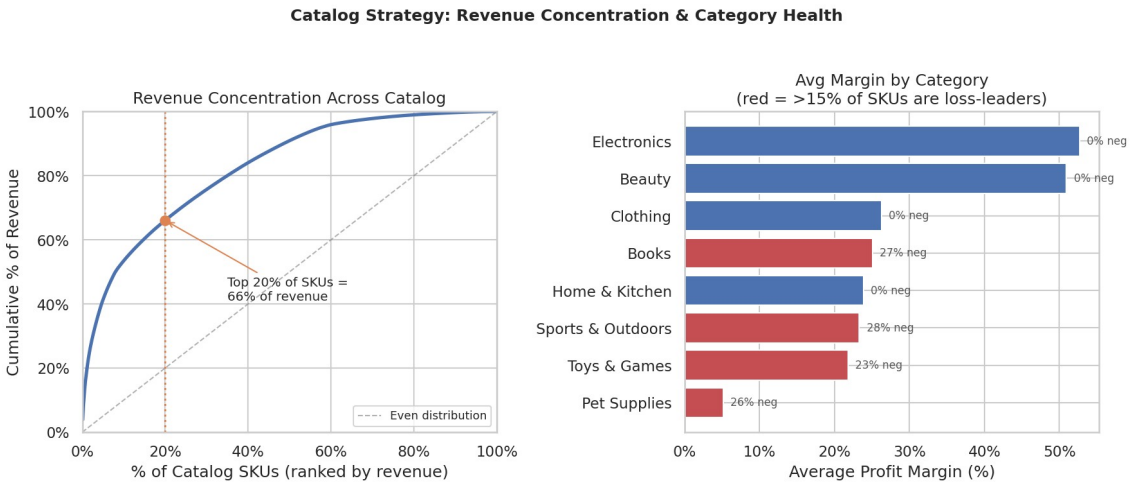


Figure 6: Revenue concentration across the SKU catalog and average margin by category

4. Summary of Recommendations

Question / Decision	Finding	Recommendation
Q1: 24/7 Staffing	Overnight = 8.5% of orders vs. 25% uniform baseline	Tiered staffing, full days, reduced overnight
Q2: Profit by Segment	VIP/Frequent drive stable profit	Invest in retention; Dormant = 0 orders, see Q5
Q3: Referral Program	No behavioral lift vs. organic buyers	Confirm A10 (shipping in discount base), run 90-day cohort analysis
Q4: Black Friday	Volume spike, small share of annual revenue	Acquisition event + Q1 follow-up campaign
Q5: Best Customers	Similar revenue across regions	Focus on channel quality + VIP frequency KPIs
Catalog Strategy	Long tail still sells; real risk is vendor continuity (53% of SKUs tied to inactive vendors)	Re-source inactive-vendor SKUs; audit Pet Supplies pricing/sourcing

5. Stakeholder Applications

Stakeholder	Goal	What the Analysis Provides
Operations	Right-size staffing without degrading SLA	Hour-by-hour volume distribution; tiered scheduling model; fulfillment SLA as guardrail
Finance & Marketing	Allocate retention spend where it generates durable return	Segment profit stability by quarter; Dormant buyer diagnosis; QoQ trend monitoring
Business Development	Assess referral ROI and restructure if ineffective	Referred vs. organic cohort comparison; cost estimate at assumed rate; 90-day cohort plan
Growth & Acquisition	Identify highest-quality acquisition channels	Regional revenue-per-buyer benchmarks; BF new-buyer conversion metric; 6-KPI tracking plan
Merchandising	Decide whether to prune or maintain catalog breadth	SKU-level revenue concentration; vendor-activity risk flag; category margin health

6. Dataset Statistics

Metric	Value
Sales rows (after dedup)	350,804
Unique buyers	47,500

Metric	Value
Vendor SKUs	1,037 across 11 files
SKU match rate (post-fix)	100%
Datetime parse coverage	> 98%
Quantity valid rate	> 99.8%
SKUs tied to a currently active vendor	47% (487 / 1,037)
Pipeline checks passing	27 / 27
Pytest tests passing	6 / 6

7. Implementation Challenges

Mixed datetime formats:

order_datetime contained three formats in a single column with no type indicator. Built a multi-pass parser that detects Unix epoch integers by regex, date-only strings by pattern, and standard timestamps by elimination, all in one pass.

Silent SKU join failure (NKDLI):

A casing inconsistency in one vendor file caused 46 SKUs to drop from the join with no error or warning. Caught by adding a pre/post SKU match-rate check; fixed by uppercasing key columns universally before sku_id construction.

Profit margin scale inconsistency:

Two of eleven vendor files stored margins on a 0–100 scale. Identified by distribution analysis (max margin ~93 vs. an expected max of ~1). Fixed with a file-aware normalization step keyed on _source_file.

Pandas dtype trap:

Initializing a column with a bare np.nan scalar locks the dtype to float64. Subsequent assignment of string values raises TypeError on strict pandas versions. Fixed by initializing with pd.Series(pd.NA, dtype='object') before assigning string content, the same underlying pattern surfaced twice in this pipeline, in two different derived columns.

Empirical region/timezone imputation:

Rather than hardcoding a census-table mapping of state → region, built the lookup empirically from rows that already had all three fields populated. This preserved the company's non-standard "Other" region bucket, which would not match any standard geographic mapping.

Pipeline: src/load.py → src/clean.py → src/features.py → src/checks.py | Analysis: notebooks/02_analysis.ipynb | Tests: tests/